

Week 4 project summary – HealthConnect Experience Lab

1. Specific problem addressed

The HealthConnect Clinic is experiencing challenges related to missed appointments, including difficulty understanding factors associated with appointment no-shows and inefficient use of appointment slots when patients fail to attend. As a Data Science intern, my main focus is to assess whether the available appointment data can support a machine-learning solution for predicting appointment no-shows. The proposed data science problem is therefore to develop a binary classification approach that predicts whether a scheduled appointment is likely to result in a No-Show or an Attended outcome. This work contributes to the broader HealthConnect project by providing a potential data-driven approach for identifying appointments that may be at greater risk of being missed.

2. Resources used

The following project resources were reviewed and used during Week 4:

- HealthConnect Appointment Dataset (HealthConnect_Appointment_Data.csv)
- HealthConnect Data Dictionary (HealthConnect_Data_Dictionary.xlsx)
- Week 4 HealthConnect Experience Lab Assignment

The appointment dataset contains fictional and anonymized appointment records covering patient characteristics, appointment details, booking information, previous appointment history, reminder information, distance to the clinic, waiting time, and appointment outcomes. The Data Dictionary was used to understand the meaning and intended use of the variables.

3. Key observations from Week 4

The initial assessment found that the dataset contains 5,000 appointment records and 18 variables. The dataset contains information that may be relevant to appointment attendance and no-show prediction, including:

- Patient demographics
- Appointment type and timing
- Booking lead time
- Previous appointment history

- Previous no-shows
- Reminder information
- Distance to the clinic
- Waiting time

Three variables contained missing values:

- reminder_channel: 1,366 missing values
- distance_to_clinic_km: 90 missing values
- waiting_time_minutes: 60 missing values

No completely duplicated records were identified during the initial assessment.

The appointment outcome variable contains three categories:

Appointment Outcome	Count	Percentage
No-Show	2,423	48.46%
Attended	2,314	46.28%
Cancelled	263	5.26%

For the proposed prediction problem, No-Show and Attended will form the initial binary classification target, while Cancelled appointments will be treated separately.

4. Proposed approach

The proposed machine learning problem is a supervised binary classification problem.

The target variable will be:

appointment_outcome

For the initial modelling approach:

- No-Show will represent the positive class.
- Attended will represent the negative class.

- Cancelled appointments will be excluded from the initial prediction target because cancellation represents a different outcome from failing to attend.

Potential input features identified from the Data Dictionary include:

- gender
- age
- age_group
- appointment_type
- appointment_day
- appointment_time
- booking_lead_days
- previous_appointments
- previous_no_shows
- reminder_sent
- reminder_channel
- distance_to_clinic_km
- waiting_time_minutes

Identifier variables such as appointment_id and patient_id are not proposed as predictive features. During the next stage, the selected data will be prepared for modelling. This is expected to include appropriate treatment of missing values, preparation of categorical variables, consideration of date-related information, and encoding of the target variable. Suitable classification models will then be investigated and compared. Model performance will be evaluated using appropriate classification metrics, including precision, recall, F1-score, and a confusion matrix. Accuracy may also be reported but will not be used as the sole evaluation measure.

5. Key considerations

Several factors may affect the proposed modelling approach.

Missing data

Missing values will require appropriate treatment during the data preparation stage. The meaning of missing values will be considered before selecting an appropriate handling method. In particular, missing values in reminder_channel require consideration because the Data Dictionary indicates that None may occur when no reminder was sent.

Data leakage

Potential data leakage must be considered when selecting features. Variables should only be used if the relevant information is available at the time a no-show prediction is made. waiting_time_minutes will therefore require further assessment before modelling.

Class Distribution

The appointment outcomes contain three categories, with No-Show and Attended representing the majority of records. The class distribution will be reassessed after the cancelled appointments are treated separately during model development.

Dataset Limitations

The dataset is fictional and anonymized. Therefore, patterns identified from the data may not fully represent real-world patient behaviour or clinic operations. Missing information and the availability of certain variables at the prediction point may also affect the final modelling approach. These considerations will be taken into account when developing and interpreting the machine learning solution.

6. Proposed Week 5 focus

The next stage will focus on preparing the dataset for machine learning and beginning model development.

The planned activities include:

1. Preparing the target variable,
2. Applying appropriate treatment to missing values,
3. Preparing the selected input features,
4. Assessing potential data leakage,
5. Splitting the data appropriately for modelling,
6. Developing and comparing suitable classification models,

7. Evaluating model performance using appropriate classification metrics.

The Week 5 activities will build directly on the problem definition and initial assessment completed during Week 4.

7. Conclusion

Week 4 established the foundation for the Data Science contribution to the HealthConnect Experience Lab. The appointment dataset was reviewed, its structure and initial data quality were assessed, and a specific machine learning problem was defined. The proposed solution is a binary classification approach for predicting appointment no-shows, using relevant appointment, booking, patient history, reminder, and accessibility-related information. The identified missing data, potential data leakage, treatment of cancelled appointments, and limitations of the synthetic dataset will be considered during the next stage of development. The Week 4 work therefore provides a structured foundation for progressing to data preparation and model development in Week 5.